

Rabbits, wildcats, pond turtles and a river that stayed: a synthesis and research agenda for trophic and hydrological rewilding in the Greater Côa Valley, Portugal

Linda Angulo Lopez, Côa Connectivity Lab; Rewilding Academic Talent Program, Rewilding Portugal

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Correspondence: Linda Angulo Lopez, coa-connectivity-lab (GitHub). ORCID: 0009-0009-5689-7148

Version: preprint draft v0.1, prepared for deposit on Zenodo. Not peer reviewed. A literature synthesis, research agenda and illustration method. It reports no new field data.

Abstract

Rewilding projects usually treat three questions separately: which predators to restore, how to recover prey, and how to restore rivers. In the Greater Côa Valley, northern Portugal, the three meet in three animals. The European rabbit (*Oryctolagus cuniculus*) is the keystone prey of the Iberian Peninsula and is now listed as Endangered in its native range. The European wildcat (*Felis silvestris*) depends on it. The Mediterranean pond turtle (*Mauremys leprosa*) links the pair to the river: an omnivore that comes through floods, drought and polluted water, provided summer pools hold water. The Côa itself is a largely free-flowing river, because a public campaign in the 1990s, “As gravuras não sabem nadar” (the engravings can’t swim), stopped the dam that would have drowned its Palaeolithic rock art. This paper brings together about 45 references on (i) predator and mesopredator cascades, (ii) the rabbit as both endangered native and invasive alien, and (iii) rewilding, the water cycle and the pond turtle. It joins them into one chain of links for the valley, from grazing and fire through rabbits and wildcats to summer pools, pond turtles and river health. From that chain it sets out seven testable hypotheses. Each hypothesis is tied to a concrete step in the open Côa connectivity workflow (Angulo Lopez, 2026a): the rabbit joins the land group as a focal species, the pond turtle joins the water group, and the wildcat model is tested against rabbit habitat. The paper also describes an illustration method: new drawings of the rabbit, the wildcat and the pond turtle made in the fine-line style of the valley’s Iron Age engravings, to be published during Global Artivism Month (1 November to 10 December 2026). None of the three animals appears in the known Côa rock-art record (Angulo Lopez, 2026b). Drawing them in the valley’s own graphic language is a way to ask which animals the next chapter of this landscape will include.

Keywords: rewilding; trophic cascades; European rabbit; European wildcat; Mediterranean pond turtle; mesopredator release; prey switching; catchment hydrology;

1. Introduction

I grew up on South Africa's Wild Coast and later worked on catchment management and post-mining land restoration in South Africa's national environment department (1997 to 2002). Those years spanned the National Water Act of 1998 (Republic of South Africa, 1998). The Act sets aside an ecological Reserve: the water a river needs to stay a living system is secured before any licence to use it is granted, and management is organised by catchment through catchment management agencies. The river holds a first claim on its own water, a move towards recognising the rights of nature and towards earth law. Under that law, rivers were never only a water question. What happened on the hillslopes, which animals grazed them and who managed them decided what reached the river. This paper brings the same catchment view to rewilding in the Greater Côa Valley.

This is not a systematic review. It is a structured synthesis of a focused set of papers, found with an AI-assisted search tool and then read and checked by hand (Section 2). I am an expert in Functional Ecology, a Geospatial Analyst and a Field Illustrator, not a mammal ecologist, a herpetologist or a hydrologist. The hypotheses in Section 6 are written so that people with that expertise can test them, or reject them.

The paper builds on a companion preprint (Angulo Lopez, 2026a). That preprint mapped land, water and air connectivity for ten focal species over a 10,230 km² study area around the Côa. It did not model any prey species. The rabbit, which most of the Côa's carnivores and large raptors depend on, was not in the model at all. Its water group also lacked a species that lives across bank and channel. That preprint named the Eurasian beaver as the clearest gap, but the beaver's native status in Portugal is uncertain, and this paper sets it aside. It works instead with the Mediterranean pond turtle, a native river animal of the Douro basin. This paper asks what those gaps cost, and whether three linked species, the rabbit, the wildcat and the pond turtle, can fill them.

Three facts frame the question. (i) The rabbit collapsed across Iberia after myxomatosis in the 1950s and rabbit haemorrhagic disease from the late 1980s, and a new virus variant (RHDV2) has spread since 2011 (Delibes-Mateos et al., 2014; Guerrero-Casado et al., 2016; Marin-Sáez et al., 2024). The Iberian lynx and the Spanish imperial eagle, both rabbit specialists, declined with it (Lees and Bell, 2008; Moreno et al., 2004). (ii) The same animal, released in Australia, Chile and elsewhere, strips vegetation and drives soil erosion (Lees and Bell, 2008; Gübelin et al., 2023; Finlayson et al., 2021). Few conservation species are so clearly both a problem and a solution. (iii) The Côa is one of the few larger Douro tributaries without a completed hydroelectric dam near its mouth. The Foz Côa dam, under construction in the early 1990s, was abandoned after the rock-art campaign (Zilhão, 2000; DeSilvey and Bartolini, 2018).

2. Approach

2.1 Literature search

The search was run in September 2026 with Consensus, an AI search engine over peer-reviewed literature, using five questions: (i) rabbits, wildcats, rewilding and river health; (ii) semi-wild herbivores for the Côa; (iii) riparian species that link water and land; (iv) the anti-dam campaign and its link to rewilding; (v) the Côa rock art, its styles and dating. Two follow-up questions, run with the same tool, covered rabbit disease in Iberia and the pond turtle as a rewilding species. The tool returned summaries with references. The references (20 exported in RIS format for question (i), plus the reference lists of the other questions) form the reading list. The summaries themselves are not reproduced.

2.2 Screening and checking

A reference was kept if it gave evidence on one of the three strands, on one of the three species, or on the Côa itself. Each claim used here is attributed to the paper it came from, not to the search tool. Where the tool's summary gave a claim that the cited paper may not support, the claim is either dropped or marked [verify] in this draft; the tag is removed once the claim is checked. Why this matters: one summary described the Mediterranean pond turtle (*Mauremys leprosa*) as herbivorous, although it is an omnivore and scavenger (Le Gal et al., 2024). Another listed a key pond turtle reintroduction paper under one author where it appears to have two, and several turtle findings came from a different species, the European pond turtle.

2.3 Limits

The reading list is small and weighted towards English-language journals. Portuguese and Spanish grey literature on rabbit restocking, wildcat surveys, pond turtle records and river management is almost certainly under-represented. Most pond turtle evidence comes from France and Spain, and some from the related European pond turtle (*Emys orbicularis*). Fix: a second pass through Portuguese sources (ICNF, university theses, the Côa Parque foundation) before v1.0.

3. Four strands of evidence

3.1 Predators and mesopredators

Where apex predators return, they can suppress smaller predators. In Spain, areas where the Iberian lynx (*Lynx pardinus*) was reintroduced held far fewer red foxes and Egyptian mongooses, and total predation pressure on rabbits by the carnivore guild fell by about half (Jiménez et al., 2019) [verify: the ca. 80% mesopredator and 55.6% guild-predation figures]. The result matters because it runs against a common fear, that restoring a predator means more predation on prey.

The effect is not guaranteed. Derham et al. (2018) find that top-down control of mesopredators depends on productivity and on landscape, and weakens in fragmented farmland. Australian rabbit-control experiments show the opposite risk: when rabbits fall

fast, feral cats and foxes switch to native prey (Finlayson et al., 2021). Risk and fix together: any rabbit or predator intervention in the Côa should be paired with monitoring of mesopredators and of alternative prey from the start, not added later.

The Côa has no lynx today. The lynx has been reintroduced in southern Portugal, in the Guadiana valley, since 2015. The Côa's mid-sized felid is the European wildcat, which the companion preprint treats as a sensitive species. Araújo and Alagador (2024) find that western Iberian reserves, the Côa included, hold fewer carnivore species than their potential. So the Côa question is narrower than the lynx literature: can a recovering wildcat population, alongside foxes, genets and raptors, hold mesopredators in check, and does it need the rabbit to do so?

3.2 The rabbit paradox

Lees and Bell (2008) call the rabbit “a conservation paradox”: an invasive alien across much of the world and an endangered native in Iberia. In its native range it feeds more than 30 predator species and digs warrens that other animals use. Restocking has been used for decades to support predators, with mixed results. In Spain, releases worked best in good habitat, in small groups and at the right season (Moreno et al., 2004). Food quality, and protein in particular, shapes how well wild populations hold up (Llobat and Marín-García, 2022). Near cities, rabbit densities follow pasture, roads and distance from forest refuges for predators (Cabrera et al., 2025).

The invasive side is the other half of the evidence, and it is useful here. In Chile's semi-arid ecosystems (Gübelin et al., 2023) and in arid Australia (Finlayson et al., 2021), rabbits show what heavy grazing and burrowing do to soil and vegetation when no predator holds them. Eldridge and Soliveres (2023) take a more balanced view: digging mammals, native or not, turn soil, capture water and seed, and can partly stand in for lost diggers. Thulin et al. (2017) add a genetic angle, asking whether dedomestication genetics could help wild rabbit recovery.

For the Côa, the lesson is that rabbit recovery is a means, not an end. A valley with more rabbits and no working predator guild is closer to Chile than to Doñana.

3.3 Rewilding, the water cycle and the pond turtle

Harvey and Henshaw (2023) review how rewilding changes the water cycle. Three mechanisms are relevant here. (i) Regrowing vegetation intercepts rain and raises soil infiltration, which lowers flood peaks. (ii) Riparian engineers such as the beaver slow flow, trap fine sediment and store water into dry periods. (iii) Reconnecting rivers with their floodplains restores storage and channel complexity. Hartig (2023) documents the return of river otters and other species as rivers in the Detroit and Windsor area recovered, a case where predator recovery followed water quality, not the reverse.

Tanentzap et al. (2023) compare passive rewilding (leaving land alone) with trophic rewilding (adding missing animals). They find trade-offs between biodiversity and ecosystem functions that depend on which path is taken [verify: summarise their result

precisely]. Svenning et al. (2024) argue that trophic rewilding is a restoration approach suited to novel conditions, where a return to any past baseline is no longer possible.

The Côa is a Mediterranean river with a hard summer low-flow season and a long fire record: 10% of the companion study area burned between 2015 and mid-2025 (Angulo Lopez, 2026a). In that setting, water retention and green riparian strips are fire and drought questions as much as biodiversity ones. They also decide which river animals can live there. The beaver, the engineer in Harvey and Henshaw's review, is set aside here (Section 1). The question becomes which native animal can use the pools the river already keeps through the summer.

The Mediterranean pond turtle (*Mauremys leprosa*) is that animal in this synthesis. It is an omnivore and scavenger, feeding on macroinvertebrates, aquatic plants and carrion (Le Gal et al., 2024). It tolerates high temperatures, low water levels and polluted water, and holds dense populations below wastewater outlets (Le Gal et al., 2023; Le Gal et al., 2024). On the Baillaury River in southern France, adults kept a stable sex ratio and population structure after a 30-year flood in 2014, although many were carried downstream for a time (Priol et al., 2024). Freshwater megafauna, turtles included, move seeds and nutrients along rivers and between bank and channel (He et al., 2024). In the related European pond turtle (*Emys orbicularis*), reintroduced animals eat invasive species, invasive crayfish among them (Kühn et al., 2025). Restoring the turtle is not simple. Released Mediterranean pond turtles did poorly in marginal and saline marshes in north-east Spain, where many animals left the release sites (Bertolero and Oro, 2009). Risk and fix together: release only into habitat with permanent summer pools, and monitor for years, not seasons.

3.4 A river that stayed

The fourth strand is historical. The campaign that stopped the Foz Côa dam in the 1990s saved the engravings and, with them, the river corridor in which they sit (Zilhão, 2000; Baptista, 2016). The valley became a World Heritage site in 1998, and an archaeological park and museum were built around the abandoned dam works. DeSilvey and Bartolini (2018) and DeSilvey (2019) show how rewilding in the valley later drew on that history. The horses in the engravings make free-roaming horses today culturally legible. DeSilvey (2019) is clear that the rock art is not used as a baseline to recreate. It supports a process of the valley "becoming wild" again, not a return to the Palaeolithic.

For a catchment view, the point is simpler. Every hydrological benefit in Section 3.3 assumes a river that can still flood, move sediment and connect. On the Côa, that condition exists because of a heritage campaign, not an ecological one.

What does a restoration plan owe to a campaign that was never about ecology?

4. One chain of links

Figure 1 joins the four strands into one chain for the Côa. [to draw: Figure 1 as a hand-drawn diagram in the illustration series, not a software chart.]

Figure 1. [to draw] Links between grazing, fire, rabbits, wildcats, summer pools, pond turtles and river health in the Greater Côa Valley.

Table 1. The chain of links, with the evidence for each and its status in the Côa.

Link	Mechanism	Evidence	Côa status
(i) Grazing → open mosaic	Semi-wild horses and cattle open scrub after land abandonment	DeSilvey and Bartolini, 2018; García-Ruiz et al., 2020; Elphick et al., 2024	Sorraia and Garrano horses, Tauros cattle released by Rewilding Portugal and ATN
(ii) Open mosaic → less intense fire	Lower fuel load, more patchy fuel	García-Ruiz et al., 2020; Bartolini et al., 2024	10% of study area burned 2015 to 2025
(iii) Open mosaic → rabbits	Short grass for food, scrub edge and soft soil for warrens	Moreno et al., 2004; Lees and Bell, 2008	No density data found [check: ICNF or Rewilding Portugal surveys]
(iv) Rabbits → predators	Keystone prey for wildcat, eagles and eagle-owl [verify: add source for eagle-owl]	Lees and Bell, 2008; Moreno et al., 2004	Wildcat present, sensitive; rabbit-dependence untested
(v) Predators → mesopredators	Top-down suppression, reduced guild predation	Jiménez et al., 2019; Derham et al., 2018	Untested; no apex felid
(vi) Summer pools → pond turtle	Permanent pools as refuge in the low-flow season; the turtle as omnivore and scavenger across bank and channel	Le Gal et al., 2024; Priol et al., 2024	Presence in the Côa basin unconfirmed; no pool survey found
(vii) Free-flowing river → all of the above	Flood pulse, sediment, corridor continuity	Harvey and Henshaw, 2023; Zilhão, 2000	Largely free-flowing; Vale de Madeira open question

Two links run backwards and matter as much. (i) Herbivores without predators can overgraze in winter shortages (Pires, 2017). (ii) Prey switching after a rabbit crash can move predation onto riparian birds, small mammals and reptiles, pond turtle nests and hatchlings among them [verify: turtle-specific evidence] (Derham et al., 2018; Finlayson et al., 2021). This backward link is what ties the rabbit to the turtle. The chain is a hypothesis, not a design.

5. Candidate species

The search surfaced candidates in three groups. This paper takes one from each as a focal species: the rabbit as prey, the wildcat as predator and the pond turtle as the river

animal. The other species in Table 2 support the chain around them. Table 2 sets each against its function, its main risk and a fix. Three further candidates are set aside: the European bison, which is poorly suited to a Mediterranean climate (Nores et al., 2024); the Eurasian beaver, whose native status in Portugal is uncertain and which conflicts with irrigation, so an evidence review should come before any position; and any species chosen to raise species counts rather than to restore a function (Fernández et al., 2017; Hale and Koprowski, 2018).

Table 2. Focal and supporting species, risks and fixes.

Species	Function	Main risk	Fix
European rabbit (focal)	Keystone prey, soil turning	Disease; overgrazing without predators	Staged releases in good habitat with disease monitoring (Moreno et al., 2004); pair with predator monitoring
European wildcat (focal)	Mid-sized predator	Hybridisation with domestic cats; persecution	Cat neutering near villages; generalise all records to 5 km
Mediterranean pond turtle (<i>Mauremys leprosa</i>) (focal)	Omnivore and scavenger across bank and channel (Le Gal et al., 2024); resilient to floods (Priol et al., 2024)	Low-flow pools drying; releases failing in marginal habitat (Bertolero and Oro, 2009)	Keep refuge pools in summer; release only into habitat with permanent water, with long-term monitoring (Stark and Galetti, 2024 on reptiles as rewilding agents)
Semi-wild horses and cattle	Grazing, fuel reduction	Overgrazing without predators	Stocking density reviewed against vegetation trends (Elphick et al., 2024)
Southwestern water vole (<i>Arvicola sapidus</i>)	Riparian prey linking river and land	Loss of wet meadows	Protect riparian grassland in grazing plans [verify: add source]
Golden and Bonelli's eagles	Aerial predators of rabbits	Power lines, disturbance	Line marking; nest buffers

Peterson et al. (2021) argue that multispecies reintroductions under deep uncertainty need a formal risk framework and staged releases. The Iberá programme in Argentina, which restored a whole guild of lost species in stages, from anteaters to jaguars, is the clearest large-scale precedent (Zamboni et al., 2017). It is also a Global South case with lessons Europe has been slow to take: it built its programme on land acquisition, local employment and a long public campaign, not on species lists alone.

6. Hypotheses and how to test them

Each hypothesis below names what would count as evidence, and the step in the Côa connectivity workflow (Angulo Lopez, 2026a) that would test it.

- i. **H1. Rabbits follow the grazing mosaic.** Rabbit density is higher where semi-wild herbivores keep an open mosaic than in abandoned scrub. Evidence: pellet or latrine counts on transects across grazed and ungrazed areas. Workflow step: add the rabbit as a land-group focal species, trained on the transect data rather than GBIF.
- ii. **H2. Wildcat habitat tracks rabbit habitat.** Modelled wildcat suitability overlaps rabbit suitability more than chance. Evidence: camera-trap detections at 5 km resolution. Workflow step: add rabbit suitability as a covariate in the land model.
- iii. **H3. Fire resets the rabbit mosaic, for better and worse.** Recently burned cells gain rabbit food but lose cover for several years. Evidence: rabbit signs by time since fire, using the MODIS fire layer already in the workflow.
- iv. **H4. Water connectivity is limited by summer flow, not by barriers alone.** Evidence: summer low-flow records at Côa gauging stations [check: SNIRH stations]. Workflow step: add a flow-permanence covariate to the water group, which the companion preprint lacks.
- v. **H5. Pond turtles persist where summer pools persist.** Pond turtle presence along the Côa and its tributaries follows the permanence of summer pools more than channel size. Evidence: visual counts or trapping at pools in August, with pool depth and duration recorded [check: ICNF permit for trapping]. Workflow step: reuse the flow-permanence covariate from H4 and add the pond turtle to the water group.
- vi. **H6. A rabbit baseline protects the turtle.** Predation on pond turtle nests and hatchlings is lower where rabbits are abundant, because predators have their main prey. Evidence: nest predation at pond-side nesting sites, compared against rabbit density from the H1 transects. Workflow step: overlay rabbit suitability (H1) with pond turtle pools (H5) to choose paired survey sites.
- vii. **H7. The rock-art corridor is also the best combined corridor.** The companion preprint found the strongest multispecies corridor along the lower Côa to the Douro. Evidence: repeat the combined surface with the rabbit and the pond turtle added and test whether the lower Côa still ranks first.

7. What this could mean for the Greater Côa Valley

- i. **A prey baseline.** No rabbit density data were found for the valley. A simple, repeated transect survey is the cheapest step with the largest return, and farmers, hunters and volunteers can run it.

- ii. **Hunting zones as partners.** Hunting zones cover 64.8% of the companion study area (Angulo Lopez, 2026a), and hunters already manage rabbits. A shared rabbit baseline gives rewilding and hunting interests one dataset to argue over, not two.
- iii. **Pond turtle: survey before release.** Pond turtle survey were found for the Côa (Angulo Lopez, 2026a). A summer survey of pools, with turtle counts, should come before any release. If releases are needed, they should go only into habitat with permanent water and be monitored for years (Bertolero and Oro, 2009).
- iv. **The free-flowing river as an asset.** The Côa's undammed lower reach is rare in the Douro basin. It is a heritage asset and a hydrological one. Funding and communication can say so together.
- v. **Learning from Iberá.** Staged, multispecies restoration with a visible public story is proven at scale in Argentina (Zamboni et al., 2017). The Côa can adopt the staging and the public story without copying the land-purchase model.
- vi. **Rabbit, wildcat and pond turtle as one package.** The three can be restored as one project, not three. The rabbit restores prey on the slopes, the wildcat holds mesopredators on the same slopes, and the pond turtle restores a consumer in the channel and its pools. They are linked through prey switching: when rabbits crash, predators turn to reptiles and other riparian prey (Derham et al., 2018; Finlayson et al., 2021), so a rabbit baseline also protects the turtle (H6). The turtle adds a condition the other two do not show: a river that keeps water through the summer (Priol et al., 2024; Le Gal et al., 2024). That is the condition the ecological Reserve in South Africa's National Water Act secures in law (Republic of South Africa, 1998).

8. Illustration method: new animals in an old line

8.1 Why

The Côa engravings record aurochs, horses, ibex and deer. The rabbit, the wildcat and the pond turtle are not known in the published corpus [verify: check the rock-art inventory (Reis, 2018) and the linked-open-data catalogue (Baptista et al., 2026) for lagomorph, felid or turtle motifs]. That absence is the starting point. The illustrations draw the three animals this paper is about in the graphic language of the valley, as new works, to ask whether they belong in its next chapter.

8.2 Reference style

The reference is the Iron Age fine-line engraving of the Côa, not the Palaeolithic figurative art. Iron Age engravings are thin incised lines, often cut with a sharp point in repeated strokes, showing warriors, horses and weapons. Rock 3 at Vermelhosa has been documented in detail by photogrammetry (Botica et al., 2023), and Botica et al. (2022) propose a set of attributes and descriptors for Iron Age motifs in the valley. Those descriptors become the drawing grammar: line width and depth, repeated or single

strokes, infill of bodies with hatching or grids, and the use of profile [verify: confirm these descriptors against Botica et al., 2022].

Source rule, as in all field notes for this project: every visual reference is labelled as either my own photograph or published documentation, with its source (Angulo Lopez, 2026a).

8.3 Steps

- i. **Study the line.** Copy three to five published Iron Age motifs by hand, in pencil, at the scale of the originals, to learn the stroke and the infill. These studies are exercises and are not published.
- ii. **Study the animal.** Draw each species from life or from reference. Rabbit: own field sketches and photographs where possible. Wildcat: published or museum references only, with no location, following the project's 5 km sensitivity rule. Pond turtle: own field sketches and photographs from C  a pools where possible, otherwise published references, labelled as such.
- iii. **Reduce to a profile.** Reduce each animal to a single profile outline in the Iron Age manner, keeping one diagnostic feature: the rabbit's ears and short tail, the wildcat's blunt, ringed tail, the pond turtle's low, oval shell and striped neck.
- iv. **Cut the line.** Make the final plate in a medium that works by incision, so that the line is cut, not drawn [choose: drypoint on plate, scratchboard, or incised fine-liner on textured paper]. Photograph my own schist from the field as a background texture only if it is clearly labelled.
- v. **Caption honestly.** Every plate is captioned as a new work "after the Iron Age fine-line engravings of the C  a Valley", with the date and my name. No plate is presented as, or could be mistaken for, an ancient motif.
- vi. **Pair with one line of evidence.** Each plate carries one sentence from this paper: the rabbit as keystone prey; the wildcat as the valley's mid-sized predator; the pond turtle as the animal that holds on in the river's summer pools.

8.4 Respect and permissions

Nothing is drawn or traced on rock, and no site is visited off permitted routes. Before publication, the plates are shared with the Funda    o C  a Parque, which manages the rock-art sites [check: contact and whether consent or courtesy notice is needed]. The published motifs used as references are cited on the gallery page.

8.5 Deliverables

- i. Three finished plates: rabbit, wildcat, pond turtle. (ii) A process sheet per species, showing line study, animal study and reduction. (iii) Captions in English, Portuguese, Spanish and French for the gallery site (Angulo Lopez, 2026b), <https://c>

oa-connectivity-lab.github.io/global-artivism/, published during Global Artivism Month (1 November to 10 December 2026) under CC BY-NC 4.0. The plates are not included in this deposit.

If the next engravers of the Côa were to draw the animals that will live there, which ones would they choose?

Data and code availability

No new data are reported. The connectivity workflow referred to in Sections 6 and 7 is the companion work: the data bundle *Côa Valley Eco-Connectivity v6 (CCL_v6)* on data.gouv.fr, <https://www.data.gouv.fr/datasets/coa-valley-eco-connectivity-v6>, and the workflow repository <https://github.com/coa-connectivity-lab/eco-connectivity-workflow>. The reading list is available as an RIS file [check: whether to include it in the deposit].

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Author contributions

Linda Angulo Lopez: conceptualisation, literature synthesis, illustration method, writing. The literature search used Consensus, an AI search tool. Drafting was assisted by an AI coding assistant (Claude, Anthropic). All claims are to be checked against the cited papers by the author before deposit.

Competing interests

None declared.

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